

Trijet cross section at large N_c (Massive case)

Marked version. Everything changed with respect to the supplied `Trijet_massive.tex` is printed in **red**; terms that had to be removed are struck through in red (~~Q^2~~). Appendix A lists every change with its equation number, page and original line number (all references are clickable).

1 Setup

The cross section for trijet production is given by

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} (2\pi) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) = \frac{1}{2} \int_{w, w'} e^{-iq \cdot (w - w')} \langle \gamma_T^i(Q, q^+, w') | \mathcal{N}_q(k_1) \mathcal{N}_{\bar{q}}(k_2) \mathcal{N}_g(k_3) | \gamma_T^i(Q, q^+, w) \rangle_{q\bar{q}g} \quad (1.1)$$

1.1 Kinematics

The virtual photon with the four momentum as $q^\mu = (q^+, -\frac{Q^2}{2q^+}, q)$. Let us define the kinematics in the following: The onshell

Component	Longitudinal momentum fraction	Transverse momentum	mass
Quark	$\zeta = \frac{k_1^+}{q^+}$	k_1	m_f
Anti Quark	$\eta = 1 - \zeta - \xi$	k_2	m_f
Gluon	$\xi = \frac{k_3^+}{q^+}$	k_3	0

denominators are $E_q(k) = \frac{k^2 + m_f^2}{2k^+}$, $E_{\bar{q}}(p) = \frac{p^2 + m_f^2}{2p^+}$, $E_g(p) = \frac{p^2}{2p^+}$.

1.2 Matrix Elements

The matrix elements that are used in this process are as follows:

1. Virtual photon splitting into a quark and anti quark pair

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta}}{2\sqrt{2q^+}} \chi_{\lambda_1}^\dagger \left[\frac{2q^\lambda}{q^+} - \sigma^\lambda \frac{\sigma \cdot p + m_f \sigma^3}{p^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda \right] \chi_{\lambda_2} \quad (1.2)$$

rewriting it using the delta function,

$$k^+ = \zeta q^+, p^+ = (1 - \zeta) q^+, \tilde{k} = k - \zeta q$$

we get

$$\left[\frac{2q^\lambda}{q^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda - \sigma^\lambda \frac{\sigma \cdot (q - k) + m_f \sigma^3}{(q^+ - k^+)} \right] = \frac{[(2\zeta - 1) \delta^{\lambda j} + i\epsilon^{\lambda j} \sigma^3] \tilde{k}^j + m_f \sigma^3 \sigma^\lambda}{\zeta (1 - \zeta) q^+}$$

where we used $\sigma^i \sigma^j = \delta^{ij} + i\epsilon^{ij} \sigma^3$ and $\sigma^\lambda \sigma^3 = -\sigma^3 \sigma^\lambda$ and

$$\chi_{\lambda_1}^\dagger [(2\zeta - 1) \delta^{\lambda j} + i\epsilon^{\lambda j} \sigma^3] \chi_{\lambda_2} = \delta_{\lambda_1 \lambda_2} [(2\zeta - 1) \delta^{\lambda j} + 2i\epsilon^{\lambda j} \lambda_1]$$

$$\chi_{\lambda_1}^\dagger \sigma^3 \sigma^\lambda \chi_{\lambda_2} = \delta_{\lambda_2, -\lambda_1} [2\lambda_1 \delta^{\lambda, 1} - i \delta^{\lambda, 2}] \equiv \mu_{\lambda_1 \lambda_2}^\lambda$$

The mass term flips the helicity ($\lambda_2 = -\lambda_1$). The matrix element defined above can be rewritten as follows:

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta} \left\{ [(2\zeta - 1) \delta^{\lambda j} + 2i\epsilon^{\lambda j} \lambda_1] \delta_{\lambda_1 \lambda_2} \tilde{k}^j + m_f \mu_{\lambda_1 \lambda_2}^\lambda \right\}}{2\sqrt{2q^+} \zeta (1 - \zeta)} \quad (1.3)$$

2. Gluon emission from anti - quark

$$\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{\alpha'}(q') | H_{\bar{q}g} | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle = -(2\pi)^6 \delta^{(3)}(k - k') \delta^{(3)}(p - p' - q') \delta_{\alpha' \alpha} \delta_{\lambda_1 \lambda_1'} \frac{g t_{\beta \beta'}^{\alpha'}}{2\sqrt{2} q'^+} \chi_{\lambda_2}^\dagger \left[\frac{2q'^{i'}}{q'^+} - \sigma^{i'} \frac{\sigma \cdot p' + m_f \sigma^3}{p'^+} - \frac{\sigma \cdot p + m_f \sigma^3}{p^+} \sigma^{i'} \right] \chi_{\lambda_2'} \quad (1.4)$$

with $p^+ = (1 - \zeta) q^+$, $p'^+ = \eta q^+$, $q'^+ = \xi q^+$ and $\tilde{p} = q' - \frac{\xi}{(1-\zeta)} p$. The terms proportional to p cancel and from here onwards we term $\omega_f = \frac{\xi m_f}{(1-\zeta)}$, **and the reduced vertex is**

$$\chi_\lambda^\dagger [\dots] \chi_{\lambda_2} = \frac{\tau_{\lambda\lambda_2}^{i'n}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^{i'}}{\xi \eta q^+}, \quad m_f \sigma^3 \sigma^{i'} \left(\frac{1}{p'^+} - \frac{1}{p^+} \right) = \frac{\xi^2 m_f}{(1-\zeta)} \frac{\sigma^3 \sigma^{i'}}{\xi \eta q^+}.$$

3. Instantaneous contribution

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(s) g_j^b(p) | H_{\gamma \rightarrow q\bar{q}g} | \gamma \rangle = -(2\pi)^3 \delta^{(3)}(q - k - p - s) \frac{g e e_f \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^b}{4 \sqrt{q^+ p^+}} \frac{\delta^{ij} + 2i \epsilon^{ij} \lambda_1}{p^+ + s^+} \quad (1.5)$$

1.3 Notations

$$\begin{aligned} \tilde{k} &= k - \zeta q, & \tilde{p} &= p - \frac{\xi}{1-\zeta} (q - k), & \tilde{Q}^2 &= \zeta(1-\zeta) Q^2 \\ \epsilon_f^2 &= \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta) Q^2 + m_f^2, & \omega_f &= \frac{\xi m_f}{(1-\zeta)}, & E_f^2 &= Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), & a &= \frac{\xi \eta}{\zeta(1-\zeta)^2} \\ \varphi_{\lambda_1 \lambda}^{ij}(\zeta) &= \delta_{\lambda_1 \lambda} [(2\zeta - 1) \delta^{ij} + 2i \epsilon^{ij} \lambda_1], & \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) &= \delta_{\lambda\lambda_2} [(2(1-\zeta) - \xi) \delta^{mn} + 2i \xi \epsilon^{mn} \lambda] \\ \mu_{\lambda\lambda'}^i &= \delta_{\lambda', -\lambda} [2\lambda \delta^{i1} - i \delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_{\lambda'}, & \sum_i \mu_{\lambda\lambda'}^i \mu_{\lambda\lambda'}^{i*} &= 2\delta_{\lambda', -\lambda} \end{aligned}$$

Here ϵ_f is the usual mass shifted photon scale. The photon vertex and the anti quark - gluon vertex are:

$$\text{photon : } \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1 \lambda}^i, \quad \text{gluon off the antiquark : } \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^m$$

The energy denominator is as follows:

$$\begin{aligned} E_q + E_{\bar{q}} - E_\gamma &= \frac{\tilde{k}^2 + \tilde{Q}^2 + m_f^2}{2\zeta(1-\zeta)q^+} = \frac{\tilde{k}^2 + \epsilon_f^2}{2\zeta(1-\zeta)q^+} \\ E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_\gamma &= \frac{(1-\zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi\eta q^+} + \frac{(\tilde{k}^2 + \epsilon_f^2)}{2\zeta(1-\zeta)q^+} = \frac{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)}{2\zeta(1-\zeta)\xi\eta q^+} \\ E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} &= \frac{p^2}{2q^+ \xi} + \frac{(q-k-p)^2 + m_f^2}{2q^+ \eta} - \frac{(q-k)^2 + m_f^2}{2q^+(1-\zeta)} = \frac{[(1-\zeta)p - \xi(q-k)]^2}{2\xi(1-\zeta)\eta q^+} + \frac{m_f^2}{2q^+} \left(\frac{1}{\eta} - \frac{1}{1-\zeta} \right) \\ &= \frac{(1-\zeta)^2 \tilde{p}^2 + \xi^2 m_f^2}{2\xi(1-\zeta)\eta q^+} \end{aligned} \quad (1.6)$$

Simplifying it we get

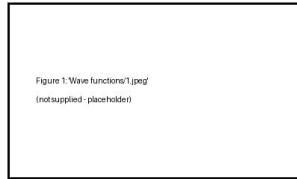
$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} = \frac{(1-\zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi\eta q^+}$$

Also, for instantaneous term involving gluon emission is given by

$$\zeta(1-\zeta)^2 \omega_f^2 + \xi\eta \epsilon_f^2 = \zeta(1-\zeta)\xi\eta E_f^2, \quad E_f^2 = Q^2 + \frac{m_f^2}{\zeta} + \frac{m_f^2}{\eta} \quad (1.7)$$

2 Outgoing component of wave function through LCPT

1. Anti quark emits a gluon before the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(1)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{\bar{q}g} | q\bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q} | H_{\gamma \rightarrow q\bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.1)$$

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1)} &= \int d^3 k'' d^3 p'' d^3 r d^3 k' d^3 p' d^3 q' d^3 k d^3 p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)| S |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_i^{a'}(q')\rangle \times \\ &\frac{\langle q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_i^{a'}(q') | H_{\bar{q}g} | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \end{aligned} \quad (2.2)$$

We will first evaluate the matrix elements without scattering (i.e., $S = 1$). With the massive vertices and energy denominators of Sec. 1, we get the following

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{1}{(2\pi)^6} \int d^2\tilde{k} d^2\tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, k) \bar{q}_{\lambda_2}^\beta(\eta q^+, q - k - p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta(1-\zeta)}{2\sqrt{\xi}(\tilde{k}^2 + \epsilon_f^2)} \frac{[\varphi_{\lambda_1\lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1\lambda}^i] [\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^m] t_{\alpha\beta}^a}{[\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)]} \quad (2.3)$$

However, Wilson lines can be obtained only if the wave function is expressed in transverse coordinates. Therefore, we perform a Fourier transform to rewrite the wave function in transverse coordinate space. Performing Fourier transform for transverse co-ordinates,

$$|q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(q - k - p) g_m^a(p)\rangle = \int_{x,y,z} e^{ik \cdot x} e^{ip \cdot z} e^{i(q-k-p) \cdot y} |q_{\lambda_1}^\alpha(x) \bar{q}_{\lambda_2}^\beta(y) g_m^a(z)\rangle \quad (2.4)$$

Rewriting k, p in terms of $\tilde{k}$ and $\tilde{p}$ respectively, we get

$$e^{i[\zeta x + \eta y + \xi z] \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i[(1-\zeta)x - \eta y - \xi z] \cdot \frac{\tilde{k}}{(1-\zeta)}} = e^{i(\zeta x + \eta y + \xi z) \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i(x - \frac{\eta y + \xi z}{(1-\zeta)}) \cdot \tilde{k}} = e^{i w \cdot q} e^{-i Y \cdot \tilde{p}} e^{i R \cdot \tilde{k}}. \quad (2.5)$$

The transverse coordinates of the photon (w) and the intermediate antiquark (y' , before emitting the gluon) are fixed by centre of energy conservation at the two vertices:

$$w = \zeta x + (1-\zeta) y' = \zeta x + \eta y + \xi z, \quad (2.6)$$

and

$$y' = \frac{\eta y + \xi z}{(1-\zeta)}. \quad (2.7)$$

The relative distances are defined as

$$R = x - y'; \quad Y = y - z$$

where we used

$$(1-\zeta)x - \eta y - \xi z = (1-\zeta)(x - y') = (1-\zeta)R$$

We will use the following Fourier transforms:

$$\int \frac{d^2k}{2\pi} \frac{d^2p}{2\pi} \frac{k^i p^j}{k^2 [ak^2 + bp^2]} e^{ik \cdot X + ip \cdot Z} = -\frac{X^i Z^j}{Z^2 [bX^2 + aZ^2]} \quad (2.8)$$

With masses the two denominators contain two different scales ϵ_f and ω_f and the transform is no longer a single Bessel function. We need four transforms, one with each helicity path. Define

$$(1-\zeta) \int \frac{d^2\tilde{k}}{2\pi} \frac{d^2\tilde{p}}{2\pi} \frac{e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \{\tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1\}}{(\tilde{k}^2 + \epsilon_f^2) [\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)]} = \left\{ R^j Y^n \mathcal{G}^{(11)}, iR^j \mathcal{G}^{(10)}, -iY^n \mathcal{G}^{(01)}, \mathcal{G}^{(00)} \right\} \quad (2.9)$$

Using Schwinger parametrization, we get

$$\frac{1}{(\tilde{k}^2 + \epsilon_f^2) [\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)]} = \frac{1}{\xi\eta} \int_0^\infty du \int_0^\infty dv e^{-(u+v)(\tilde{k}^2 + \epsilon_f^2) - \frac{v}{a}(\tilde{p}^2 + \omega_f^2)} \quad (2.10)$$

where $a = \frac{\xi\eta}{[\zeta(1-\zeta)^2]}$, K_ν is the Modified Bessel function of the second kind. Using the Gaussian integrals and including massive terms, we define the following notations, with x as a Feynman parameter.

$$\mathcal{G}^{(\alpha\beta)}(R, Y) = \frac{a^\beta}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{A_x}{\Delta_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{A_x \Delta_x}), \alpha, \beta \in \{0, 1\}$$

$$A_x = \epsilon_f^2 + \frac{x\omega_f^2}{a} = \zeta(1-\zeta)Q^2 + m_f^2 \left(1 + \frac{x\zeta\xi}{\eta} \right)$$

$$\Delta_x = R^2 + \frac{aY^2}{x}$$

Explicitly

$$\mathcal{G}^{(11)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \frac{A_x}{\Delta_x} K_2(\sqrt{A_x \Delta_x}) \quad (2.11)$$

$$\mathcal{G}^{(10)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_1 \left(\sqrt{A_x \Delta_x} \right) \quad (2.12)$$

$$\mathcal{G}^{(01)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}} K_1 \left(\sqrt{A_x \Delta_x} \right) \quad (2.13)$$

$$\mathcal{G}^{(00)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_0 \left(\sqrt{A_x \Delta_x} \right) \quad (2.14)$$

Now, implementing all these steps in the outgoing component of wave function, we get

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta t_{\alpha\beta}^a}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}] \quad (2.15)$$

where

$$\begin{aligned} \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}] = \sum_{\lambda} \left[\varphi_{\lambda_1\lambda}^{ij}(\zeta) \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{G}^{(11)} + i \xi \omega_f \varphi_{\lambda_1\lambda}^{ij}(\zeta) \mu_{\lambda\lambda_2}^m R^j \mathcal{G}^{(10)} \right. \\ \left. - i m_f \mu_{\lambda_1\lambda}^i \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{G}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1\lambda}^i \mu_{\lambda\lambda_2}^m \mathcal{G}^{(00)} \right] \end{aligned} \quad (2.16)$$

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta(1-\zeta) R^2 + \frac{\xi \eta Y^2}{(1-\zeta)} = \zeta \eta (x-y)^2 + \zeta \xi (x-z)^2 + \eta \xi (y-z)^2$$

$$\mu_{\lambda\lambda'}^i = \delta_{\lambda', -\lambda} [2\lambda \delta^{i1} - i\delta^{i2}] = \chi_{\lambda}^\dagger \sigma^3 \sigma^i \chi_{\lambda'}$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta) Q^2 + m_f^2, \quad \omega_f = \frac{\xi m_f}{(1-\zeta)}, \quad E_f^2 = Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), \quad a = \frac{\xi \eta}{\zeta(1-\zeta)^2}$$

$$\varphi_{\lambda_1\lambda}^{ij}(\zeta) = \delta_{\lambda_1\lambda} [(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1], \quad \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) = \delta_{\lambda\lambda_2} [(2(1-\zeta)-\xi)\delta^{mn} + 2i\xi\epsilon^{mn}\lambda]$$

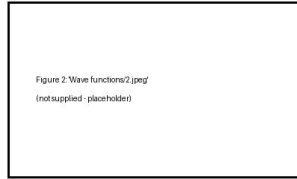
and

$$c = \zeta x + \eta y + \xi z$$

is the centre of energy of trijet system. Attaching Wilson lines, we get

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}](R, Y) \\ [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.17)$$

2. Anti quark emits a gluon after the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(2)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{\bar{q}g} | q \bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q \bar{q} | H_{\gamma \rightarrow q \bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.18)$$

$$|\gamma\rangle_{q\bar{q}g}^{(2)} = \int d^3 k'' d^3 p'' d^3 r d^3 k' d^3 p' d^3 k d^3 p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r) | H_{\bar{q}g} | q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') \rangle}{E_q(k'') + E_{\bar{q}}(p'') - E_q(k') - E_{\bar{q}}(p') - E_g(r)} \times \\ \langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') | S | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q \bar{q}} | \gamma_\lambda(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.19)$$

The only difference between the first and the second wave function component is the energy denominator and the Wilson line structure. The changes are as follows:

$$\frac{1}{E_{q\bar{q}g} - E_\gamma} \rightarrow \frac{1}{E_{q\bar{q}} - E_{q\bar{q}g}}$$

Equivalently,

$$\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \epsilon_f^2)} \rightarrow -\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2)}$$

Also, the Fourier transform which we will be using here is the following:

$$\int \frac{d^2 \tilde{k}}{2\pi} e^{i\tilde{k} \cdot R} \frac{\{\tilde{k}^j, 1\}}{\tilde{k}^2 + \epsilon_f^2} = \left\{ \frac{i R^j}{R} \epsilon_f K_1(\epsilon_f R), K_0(\epsilon_f R) \right\}$$

and

$$\int \frac{d^2 \tilde{p}}{2\pi} e^{-i\tilde{p} \cdot Y} \frac{\{\tilde{p}^n, 1\}}{(\tilde{p}^2 + \omega_f^2)} = \left\{ -\frac{i Y^n}{Y} \omega_f K_1(\omega_f Y), K_0(\omega_f Y) \right\}$$

With the **same** normalisation, we get

$$(1 - \zeta) \int \frac{d^2 \tilde{k}}{2\pi} \frac{d^2 \tilde{p}}{2\pi} e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \frac{\left\{ \tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1 \right\}}{(\tilde{k}^2 + \epsilon_f^2) \zeta (1 - \zeta)^2 (\tilde{p}^2 + \omega_f^2)} = \left\{ R^j Y^n \mathcal{A}^{(11)}, i R^j \mathcal{A}^{(10)}, -i Y^n \mathcal{A}^{(01)}, \mathcal{A}^{(00)} \right\} \quad (2.20)$$

where:

$$\begin{aligned} \mathcal{A}^{(11)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(10)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R} K_0(\omega_f Y) \\ \mathcal{A}^{(01)} &= \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(00)} &= \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)} K_0(\omega_f Y) \end{aligned}$$

The Wilson line will not have any component of gluon because the gluons are not getting scattered as the gluon is emitted after the scattering.

$$[U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \rightarrow [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta}$$

Substituting all these changes in eq. (2.17), we get

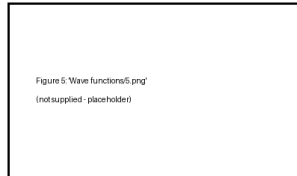
$$\begin{aligned} |\gamma(w)\rangle_{q\bar{q}g}^{(2)} &= + \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \\ &\Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}](R, Y) \times [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta} \end{aligned} \quad (2.21)$$

where

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] &= \sum_{\lambda} \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{A}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{A}^{(10)} \right. \\ &\quad \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{A}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{A}^{(00)} \right] \end{aligned} \quad (2.22)$$

In the massless case Eq. (2.21) follows from Eq. (2.17) by the formal substitution $(b) = (a)|_{y,z \rightarrow y'}$ of the paper (keeping the prefactor $1/Y^2$). With masses this is no longer literally true: the functions $\mathcal{A}^{(\alpha\beta)}$ are different functions from $\mathcal{G}^{(\alpha\beta)}$ (after the shock wave the gluon is emitted from an on-shell antiquark and sees only the dead-cone scale ω_f). We keep the label $[y, z \rightarrow y']$ for this term, meaning: Wilson lines at y' and $\mathcal{G} \rightarrow \mathcal{A}$. The short-distance behaviour is the same, $Y^2 \mathcal{G}^{(11)}, Y^2 \mathcal{A}^{(11)} \xrightarrow{Y \rightarrow 0} \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R}, Y^2 \mathcal{G}^{(01)}, Y^2 \mathcal{A}^{(01)} \xrightarrow{Y \rightarrow 0} \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)}$, and $\mathcal{G}^{(10)} - \mathcal{A}^{(10)}, \mathcal{G}^{(00)} - \mathcal{A}^{(00)}$ stay finite, so the UV cancellation between Eq. (2.17) and Eq. (2.21) works as in the massless case.

3. **Instantaneous** contribution (Anti quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.23)$$

Using the relevant matrix we get (In momentum space - without Wilson lines)

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} &= + \frac{1}{(2\pi)^6} \int d^2 \tilde{k} d^2 \tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, k) \bar{q}_{\lambda_2}^\beta(\eta q^+, q - k - p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta}{2\sqrt{\xi}} \\ &\quad \frac{(\delta^{im} + 2i \epsilon^{im} \lambda_1) \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^a}{\tilde{k}^2 + \epsilon_f^2 + \frac{\zeta(1-\zeta)^2}{\xi \eta} (\tilde{p}^2 + \omega_f^2)} \end{aligned} \quad (2.24)$$

Performing Fourier transform and inserting $S - 1$ component, we get the following

$$(1 - \zeta) \int \frac{d^2 \tilde{k}}{2\pi} \frac{d^2 \tilde{p}}{2\pi} e^{i \tilde{k} \cdot R} e^{-i \tilde{p} \cdot Y} \frac{1}{\zeta (1 - \zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \epsilon_f^2)} = \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} = \mathcal{G}_E \quad (2.25)$$

The outgoing component of wave function is obtained as follows:

$$|\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} = + \frac{e e_f g q^+}{2 (2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta \eta \sqrt{\xi}}{(1 - \zeta)} \delta^{(2)}(w - c) (\delta^{im} + 2i \epsilon^{im} \lambda_1) \delta_{\lambda_1 \lambda_2} \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.26)$$

This has the same Wilson line structure as the regular i.e., Eq. (2.17). **There is no instantaneous term after the shock wave (the gluon emission after the shock wave is a single vertex), so the instantaneous term has no $[y, z \rightarrow y']$ counterpart (in the massless note this was expressed by the factor $\frac{Y^2}{R \cdot Y}$ of the effective vertex, which vanishes for $y, z \rightarrow y'$).** Now, we have all the wave functions corresponding to anti quark emission. All four anti quark emission can be written as a single expression with an effective vertex. The effective vertex is written as

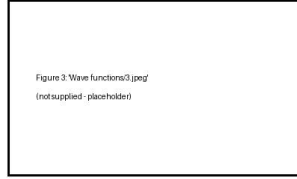
$$\mathcal{M}_{1\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}](R, Y) - \delta_{\lambda_1\lambda_2} \frac{\xi \eta}{(1 - \zeta)} (\delta^{im} + 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (2.27)$$

$$\mathcal{M}_{2\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{A}](R, Y) \quad (2.28)$$

All four anti quark emission are then written using a single expression:

$$|\gamma^i(w)\rangle_{q\bar{q}g} = - \frac{e e_f g q^+}{2 (2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w - c) \left[\mathcal{M}_{1\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} - \mathcal{M}_{2\lambda_1\lambda_2}^{im} [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta} \right] \quad (2.29)$$

4. Quark emits a gluon before the shock wave



When we move from anti quark emission to quark emission, we will have to perform the following changes from anti quark wave functions:

- $\zeta \leftrightarrow \eta$
- $x \leftrightarrow y$
- $\alpha \leftrightarrow \beta$
- $\lambda_1 \leftrightarrow \lambda_2$
- Change overall sign
- $y' \rightarrow x' = \frac{\zeta x + \xi z}{\zeta + \xi}$
- $R \rightarrow y - x' = -\tilde{R}$
- $Y \rightarrow x - z = X$
- $\varphi_{\lambda_1\lambda}^{ij}(\zeta) \rightarrow \varphi_{\lambda_2\lambda}^{ij*}(\eta) = -\varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi)$
- $\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \rightarrow \tau_{\lambda\lambda_1}^{mn*}(\xi, \zeta) = \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta)$
- $D(x, y, z, \zeta, \xi) \rightarrow D(y, x, z, \eta, \xi)$
- **after the shock wave (massless limit):** $\mathcal{D} \rightarrow \sqrt{\eta(1 - \eta)} \tilde{R}$, i.e. $\tilde{Q} \rightarrow \sqrt{\eta(1 - \eta)} Q$
- $m_f \mu_{\lambda_1\lambda}^i \rightarrow m_f \mu_{\lambda\lambda_2}^i$
- $\xi \omega_f \mu_{\lambda\lambda_2}^m \rightarrow \xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m$ with $\tilde{\omega}_f = \frac{\xi m_f}{(1 - \eta)}$
- $\epsilon_f^2 \rightarrow \tilde{\epsilon}_f^2 = \eta(1 - \eta) Q^2 + m_f^2$, $a \rightarrow \tilde{a} = \frac{\xi \zeta}{\eta(1 - \eta)^2}$
- $A_x \rightarrow \tilde{A}_x = \eta(1 - \eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta}\right)$ and $\Delta_x \rightarrow \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$ and $\mathcal{A}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$ also, $\zeta(1 - \zeta) \rightarrow \eta(1 - \eta)$

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{qg} |q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q}| H_{\gamma\rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.30)$$

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3q' d^3k d^3p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1'}^{\alpha'}(k'') \bar{q}_{\lambda_2'}^{\beta'}(p'') g_j^b(r)| S |q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{a'}(q')\rangle \times \frac{\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{a'}(q')| H_{qg} |q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)\rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)| H_{\gamma\rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.31)$$

$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(3)} = \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \left[\tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \right] \quad (2.32)$$

with

$$\tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{G}}] = \sum_\lambda \left[\tau_{\lambda_1\lambda}^{nm}(\xi, \zeta) \varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{G}}^{(11)} + i\xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m \varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j \tilde{\mathcal{G}}^{(10)} - im_f \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta) \mu_{\lambda\lambda_2}^i X^n \tilde{\mathcal{G}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m \mu_{\lambda\lambda_2}^i \tilde{\mathcal{G}}^{(00)} \right] \quad (2.33)$$

where

$$\tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X) = \frac{\tilde{a}^\beta}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{\tilde{A}_x}{\tilde{\Delta}_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{\tilde{A}_x \tilde{\Delta}_x}), \alpha, \beta \in \{0, 1\}$$

$$\tilde{A}_x = \tilde{\epsilon}_f^2 + \frac{x \tilde{\omega}_f^2}{\tilde{a}} = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right)$$

$$\tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$$

Explicitly

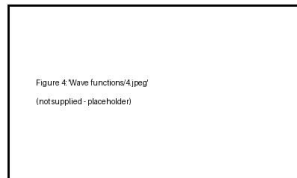
$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \frac{\tilde{A}_x}{\tilde{\Delta}_x} K_2 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.34)$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.35)$$

$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.36)$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_0 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.37)$$

5. Quark emits a gluon after the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(4)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{qg} |q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q\bar{q}| H_{\gamma\rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.38)$$

$$|\gamma\rangle_{q\bar{q}g}^{(4)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3k d^3p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1'}^{\alpha'}(k'') \bar{q}_{\lambda_2'}^{\beta'}(p'') g_j^b(r)| H_{qg} |q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p')\rangle}{E_q(k') + E_{\bar{q}}(p') - E_q(k'') - E_{\bar{q}}(p'') - E_g(r)} \times \frac{\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p')| S |q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)| H_{\gamma\rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.39)$$

Substituting the relations we get

$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(4)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \times [t^a V(x') V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.40)$$

where

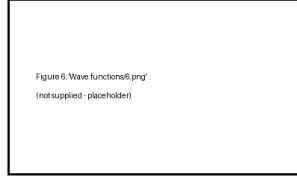
$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (2.41)$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} K_0(\tilde{\omega}_f X) \quad (2.42)$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (2.43)$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} K_0(\tilde{\omega}_f X) \quad (2.44)$$

6. Instantaneous contribution (Quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle_{q\bar{q}g}^{(3 \text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.45)$$

Using the transformations from anti quark to quark with the insertions of Wilson lines, we get

$$|\gamma\rangle_{q\bar{q}g}^{(3 \text{ inst.})} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta \eta \sqrt{\xi}}{(1-\eta)} \delta^{(2)}(w-c) (\delta^{im} - 2i \epsilon^{im} \lambda_2) \delta_{\lambda_1 \lambda_2} \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.46)$$

Now, we will write the quark emissions as how we wrote the anti quark section. We will use the effective vertex formalism.

$$|\gamma^i(w)\rangle_{q\bar{q}g} = \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \left[\tilde{\mathcal{M}}_{1\lambda_1 \lambda_2}^{im}(x, y, z, \eta, \xi) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} - \tilde{\mathcal{M}}_{2\lambda_1 \lambda_2}^{im}[t^a V(x') V^\dagger(y) - t^a]_{\alpha\beta} \right] \quad (2.47)$$

with the effective vertex

$$\tilde{\mathcal{M}}_{1\lambda_1 \lambda_2}^{im}(x, y, z, \eta, \xi) = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) - \delta_{\lambda_1 \lambda_2} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (2.48)$$

$$\tilde{\mathcal{M}}_{2\lambda_1 \lambda_2}^{im}(x, y, z, \eta, \xi) = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \quad (2.49)$$

3 Trijet production cross section

3.1 Direct contributions - Anti quark contribution

$$\frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} = \frac{e^2 g^2 \zeta^2 \cancel{Q^2} C_F N_c}{8(2\pi)^{12} \xi q^{+2}} \left(\sum_f \cancel{e_f^2} \right) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \left[\mathcal{H}[\mathcal{G}; \bar{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}[\mathcal{A}; \bar{\mathcal{G}}] \mathcal{W}(x, \textcolor{red}{y}', y', \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}[\mathcal{G}; \bar{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{H}[\mathcal{A}; \bar{\mathcal{A}}] \mathcal{W}(x, \textcolor{red}{y}', y', \bar{x}, \bar{y}', \bar{y}') \right] \quad (3.1)$$

where

$$\mathcal{H}[F; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l,r} \mathcal{M}_{\lambda_1 \lambda_2}^{lr}[F](x, y, z) \mathcal{M}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.2)$$

- Regular $\times$ Regular

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{reg.}}[F; \bar{F}] = & R^i Y^m \bar{R}^j \bar{Y}^n \left\{ 8 [\eta^2 + (1 - \zeta)^2] [\zeta^2 + (1 - \zeta)^2] \delta^{ij} \delta^{mn} - 8\xi(2\zeta - 1)(2(1 - \zeta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ & + 8 m_f^2 [\eta^2 + (1 - \zeta)^2] (Y \cdot \bar{Y}) F^{(01)} \bar{F}^{(01)} + 8 \xi^2 \omega_f^2 [\zeta^2 + (1 - \zeta)^2] (R \cdot \bar{R}) F^{(10)} \bar{F}^{(10)} \\ & + 8 m_f^2 \xi^2 \omega_f^2 F^{(00)} \bar{F}^{(00)} \\ & - 8\zeta \eta m_f \xi \omega_f \left[(R \cdot Y) F^{(11)} \bar{F}^{(00)} + (\bar{R} \cdot \bar{Y}) F^{(00)} \bar{F}^{(11)} + (R \cdot \bar{Y}) F^{(10)} \bar{F}^{(01)} + (Y \cdot \bar{R}) F^{(01)} \bar{F}^{(10)} \right] \end{aligned} \quad (3.3)$$

- Regular $\times$ Instantaneous + h.c.

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{inst.}}[F; \bar{F}] = & -\frac{8\zeta \xi \eta^2}{(1 - \zeta)} \left[(R \cdot Y) F^{(11)} \bar{\mathcal{G}}_E + (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \right] \\ & + \frac{8\xi^2 \eta m_f \omega_f}{(1 - \zeta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \end{aligned} \quad (3.4)$$

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst.} \times \text{inst.}}[F; \bar{F}] = 8 \left[\frac{\xi \eta}{(1 - \zeta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.5)$$

where:

$$\mathcal{M}_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] \equiv \mathcal{M}_{1\lambda_1 \lambda_2}^{im} = \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}](R, Y) - \delta_{\lambda_1 \lambda_2} \frac{\xi \eta}{(1 - \zeta)} (\delta^{im} + 2i\epsilon^{im} \lambda_1) \mathcal{G}_E \quad (3.6)$$

$$\mathcal{M}_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] \equiv \mathcal{M}_{2\lambda_1 \lambda_2}^{im} = \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}](R, Y) \quad (3.7)$$

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] = & \sum_{\lambda} \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{G}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{G}^{(10)} \right. \\ & \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{G}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{G}^{(00)} \right] \end{aligned} \quad (3.8)$$

Explicitly:

$$\begin{aligned} \mathcal{G}^{(\alpha\beta)}(R, Y) = & \frac{a^\beta}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{A_x}{\Delta_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{A_x \Delta_x}), \alpha, \beta \in \{0, 1\} \\ A_x = & \epsilon_f^2 + \frac{x \omega_f^2}{a} = \zeta(1 - \zeta) Q^2 + m_f^2 \left(1 + \frac{x \zeta \xi}{\eta} \right) \\ \Delta_x = & R^2 + \frac{a Y^2}{x} \end{aligned}$$

$$\mathcal{G}^{(11)} = \frac{a}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x^2} \frac{A_x}{\Delta_x} K_2(\sqrt{A_x \Delta_x}) \quad (3.9)$$

$$\mathcal{G}^{(10)} = \frac{1}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (3.10)$$

$$\mathcal{G}^{(01)} = \frac{a}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (3.11)$$

$$\mathcal{G}^{(00)} = \frac{1}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_0(\sqrt{A_x \Delta_x}) \quad (3.12)$$

and

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] = & \sum_{\lambda} \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{A}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{A}^{(10)} \right. \\ & \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{A}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{A}^{(00)} \right] \end{aligned} \quad (3.13)$$

Explicitly:

$$\begin{aligned}\mathcal{A}^{(11)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta) R} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(10)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta) R} K_0(\omega_f Y) \\ \mathcal{A}^{(01)} &= \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(00)} &= \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)} K_0(\omega_f Y)\end{aligned}$$

Finally:

$$\mathcal{G}_E = \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}}$$

where:

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta(1-\zeta) R^2 + \frac{\xi \eta Y^2}{(1-\zeta)} = \zeta \eta (x-y)^2 + \zeta \xi (x-z)^2 + \eta \xi (y-z)^2$$

$$\mu_{\lambda\lambda'}^i = \delta_{\lambda', -\lambda} [2\lambda \delta^{i1} - i\delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_{\lambda'}$$

$$\begin{aligned}\epsilon_f^2 &= \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta) Q^2 + m_f^2, & \omega_f &= \frac{\xi m_f}{(1-\zeta)}, & E_f^2 &= Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), & a &= \frac{\xi \eta}{\zeta(1-\zeta)^2} \\ \varphi_{\lambda_1 \lambda}^{ij}(\zeta) &= \delta_{\lambda_1 \lambda} [(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1], & \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) &= \delta_{\lambda\lambda_2} [(2(1-\zeta)-\xi)\delta^{mn} + 2i\xi\epsilon^{mn}\lambda]\end{aligned}$$

and

$$c = \zeta x + \eta y + \xi z$$

is the centre of energy of trijet system. $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.14)$$

At large N_c limit, we get

$$\begin{aligned}S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) &= Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) \\ S_{q\bar{q}g}(x, y, z) &= S(x, z) S(z, y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) &= S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})\end{aligned}$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - S(x, z) S(z, y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.15)$$

At large N_c limit, we get

$$\begin{aligned}S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) &= Q(x, y', \bar{z}, \bar{x}) Q(y', y', \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) S(\bar{y}, \bar{z}) \\ S_{q\bar{q}g}(x, y', y') &= S(x, y') S(y', y') = S(x, y') \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) &= S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})\end{aligned}$$

$$\boxed{\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) S(\bar{y}, \bar{z}) - S(x, y') - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.16)$$

At large N_c limit, we get

$$\begin{aligned}S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') &= Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) \\ S_{q\bar{q}g}(x, y, z) &= S(x, z) S(z, y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') &= S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})\end{aligned}$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) - S(x, z) S(z, y) - S(\bar{y}', \bar{x}) + 1}$$

- $\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.17)$$

At large N_c limit, we get

$$\begin{aligned}S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') &= Q(x, y', \bar{y}', \bar{x}) Q(y', y', \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x}) \\ S_{q\bar{q}g}(x, y', y') &= S(x, y') S(y', y') = S(x, y') \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') &= S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})\end{aligned}$$

$$\boxed{\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x}) - S(x, y') - S(\bar{y}', \bar{x}) + 1}$$

3.2 Direct contribution - Quark contributions

The following transformations are made from the anti quark sector to quark sector:

$$\zeta \rightarrow \eta, R \rightarrow \tilde{R}, Y \rightarrow X, x \leftrightarrow y$$

and

$$\epsilon_f \rightarrow \tilde{\epsilon}_f, \omega_f \rightarrow \tilde{\omega}_f, \mathcal{G} \rightarrow \tilde{\mathcal{G}}, \mathcal{A} \rightarrow \tilde{\mathcal{A}}$$

$$\frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} = \frac{e^2 g^2 \eta^2 \cancel{Q^2} C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \left(\sum_f \cancel{e_f^2} \right) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \left[\tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') + \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') \right] \quad (3.18)$$

where:

$$\tilde{\mathcal{H}}[F; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l, r} \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr}[F](x, y, z) \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.19)$$

- Regular $\times$ Regular

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{reg.} \times \text{reg.}} = & \tilde{R}^i X^m \tilde{R}^j \bar{X}^n \left\{ 8 [\zeta^2 + (1-\eta)^2] [\eta^2 + (1-\eta)^2] \delta^{ij} \delta^{mn} - 8\xi (2\eta - 1) (2(1-\eta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ & + 8m_f^2 [\zeta^2 + (1-\eta)^2] (X \cdot \bar{X}) F^{(01)} \bar{F}^{(01)} + 8\xi^2 \tilde{\omega}_f^2 [\eta^2 + (1-\eta)^2] (\tilde{R} \cdot \bar{R}) F^{(10)} \bar{F}^{(10)} + 8m_f^2 \xi^2 \tilde{\omega}_f^2 F^{(00)} \bar{F}^{(00)} \\ & + 8\zeta \eta m_f \xi \tilde{\omega}_f \left[(\tilde{R} \cdot X) F^{(11)} \bar{F}^{(00)} + (\bar{R} \cdot \bar{X}) F^{(00)} \bar{F}^{(11)} + (\tilde{R} \cdot \bar{X}) F^{(10)} \bar{F}^{(01)} + (X \cdot \bar{R}) F^{(01)} \bar{F}^{(10)} \right] \end{aligned} \quad (3.20)$$

- Regular $\times$ Instantaneous + h.c.

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{reg.} \times \text{inst.}} = & \frac{8\zeta^2 \xi \eta}{(1-\eta)} \left[(\tilde{R} \cdot X) F^{(11)} \bar{\mathcal{G}}_E + (\bar{R} \cdot \bar{X}) \mathcal{G}_E \bar{F}^{(11)} \right] \\ & + \frac{8\xi^2 \zeta m_f \tilde{\omega}_f}{(1-\eta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \end{aligned} \quad (3.21)$$

- Instantaneous $\times$ Instantaneous

$$\tilde{\mathcal{H}}_{\text{inst.} \times \text{inst.}} = 8 \left[\frac{\xi \zeta}{(1-\eta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.22)$$

where:

$$\tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}] \equiv \tilde{\mathcal{M}}_{1\lambda_1 \lambda_2}^{im} = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) - \delta_{\lambda_1 \lambda_2} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (3.23)$$

$$\tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}] \equiv \tilde{\mathcal{M}}_{2\lambda_1 \lambda_2}^{im} = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \quad (3.24)$$

with

$$\begin{aligned} \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}] = & \sum_{\lambda} \left[\tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{G}}^{(11)} + i\xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j \tilde{\mathcal{G}}^{(10)} \right. \\ & \left. - i m_f \tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \mu_{\lambda \lambda_2}^i X^n \tilde{\mathcal{G}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \mu_{\lambda \lambda_2}^i \tilde{\mathcal{G}}^{(00)} \right] \end{aligned} \quad (3.25)$$

where

$$\begin{aligned} \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X) = & \frac{\tilde{a}^\beta}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{\tilde{A}_x}{\tilde{\Delta}_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{\tilde{A}_x \tilde{\Delta}_x}), \alpha, \beta \in \{0, 1\} \\ \tilde{A}_x = & \tilde{\epsilon}_f^2 + \frac{x \tilde{\omega}_f^2}{\tilde{a}} = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right) \\ \tilde{\Delta}_x = & \tilde{R}^2 + \frac{\tilde{a} X^2}{x} \end{aligned}$$

Explicitly

$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \frac{\tilde{A}_x}{\tilde{\Delta}_x} K_2 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.26)$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.27)$$

$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.28)$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_0 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.29)$$

and

$$\begin{aligned} \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}] = \sum_{\lambda} \left[\tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{A}}^{(11)} + i\xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j \tilde{\mathcal{A}}^{(10)} \right. \\ \left. - im_f \tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \mu_{\lambda \lambda_2}^i X^n \tilde{\mathcal{A}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \mu_{\lambda \lambda_2}^i \tilde{\mathcal{A}}^{(00)} \right] \end{aligned} \quad (3.30)$$

Explicitly:

$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (3.31)$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} K_0(\tilde{\omega}_f X) \quad (3.32)$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (3.33)$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} K_0(\tilde{\omega}_f X) \quad (3.34)$$

and

- $\xi \omega_f \mu_{\lambda \lambda_2}^m \rightarrow \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m$ with $\tilde{\omega}_f = \frac{\xi m_f}{(1-\eta)}$
- $\epsilon_f^2 \rightarrow \tilde{\epsilon}_f^2 = \eta(1-\eta) Q^2 + m_f^2$, $a \rightarrow \tilde{a} = \frac{\xi \zeta}{\eta(1-\eta)^2}$
- $A_x \rightarrow \tilde{A}_x = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right)$ and $\Delta_x \rightarrow \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$ and $\mathcal{A}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$ also, $\zeta(1-\zeta) \rightarrow \eta(1-\eta)$

$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.35)$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z}) - S(y, z) S(z, x) - S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z}) + 1$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.36)$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) Q(x', x', \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$\boxed{\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) S(\bar{x}, \bar{z}) - S(y, x') - S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z}) + 1}$$

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\ q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.37)$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) Q(z, x, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) S(z, x)$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

$$\boxed{\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) S(z, x) - S(y, z) S(z, x) - S(\bar{x}', \bar{y}) + 1}$$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.38)$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y}) Q(x', x', \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

$$\boxed{\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y}) - S(y, x') - S(\bar{x}', \bar{y}) + 1}$$

3.3 Interference contributions

$$\begin{aligned} \frac{d^3 N^{\text{cross}}}{d^3 k_1 d^3 k_2 d^3 k_3} &= -\frac{e^2 g^2 \eta \zeta \cancel{Q^2} C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \left(\sum_f \cancel{e_f^2} \right) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \times 2 \Re \int_{x, y, z, \bar{x}, \bar{y}, \bar{z}} e^{-i k_1 \cdot (x - \bar{x})} e^{-i k_2 \cdot (y - \bar{y})} e^{-i k_3 \cdot (z - \bar{z})} \sum_f \cancel{e_f^2} \\ &\left[\mathcal{H}^x[\tilde{\mathcal{G}}, \tilde{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}^x[\tilde{\mathcal{A}}, \tilde{\mathcal{G}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}^x[\tilde{\mathcal{G}}, \tilde{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') \right. \\ &\left. + \mathcal{H}^x[\tilde{\mathcal{A}}, \tilde{\mathcal{A}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') \right] \end{aligned} \quad (3.39)$$

where

$$\mathcal{H}^x[\tilde{F}, \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l, r} \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr}[\tilde{F}](x, y, z) \mathcal{M}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.40)$$

- Regular $\times$ Regular

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{reg.}}^x &= \tilde{R}^i X^m \tilde{R}^j \bar{Y}^n \left\{ 8[1 - \xi - 2\zeta\eta] [\xi(1 - \xi) + 2\zeta\eta] \delta^{ij} \delta^{mn} - 8\xi[1 - 2\zeta - \xi]^2 \epsilon^{ij} \epsilon^{mn} \right\} \tilde{F}^{(11)} \bar{F}^{(11)} \\ &+ 8m_f^2 (\xi + 2\zeta\eta) (X \cdot \bar{Y}) \tilde{F}^{(01)} \bar{F}^{(01)} - 8(\xi + 2\eta\zeta) \xi^2 \tilde{\omega}_f \omega_f (\tilde{R} \cdot \bar{R}) \tilde{F}^{(10)} \bar{F}^{(10)} - 8(\zeta + \xi)^2 m_f \xi \omega_f (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{F}^{(00)} \\ &+ 8(1 - \zeta)^2 m_f \xi \tilde{\omega}_f (\bar{R} \cdot \bar{Y}) \tilde{F}^{(00)} \bar{F}^{(11)} + 8\eta^2 m_f \xi \tilde{\omega}_f (\tilde{R} \cdot \bar{Y}) \tilde{F}^{(10)} \bar{F}^{(01)} - 8\zeta^2 m_f \xi \omega_f (X \cdot \bar{R}) \tilde{F}^{(01)} \bar{F}^{(10)} \end{aligned} \quad (3.41)$$

- Regular $\times$ Instantaneous + h.c.

$$\mathcal{H}_{\text{reg.} \times \text{inst.}}^x = -\frac{8\xi\eta(\xi + \zeta)^2}{(1 - \zeta)} (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{\mathcal{G}}_E + \frac{8\xi\zeta(1 - \zeta)^2}{(\xi + \zeta)} (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \quad (3.42)$$

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst.} \times \text{inst.}}^x = \frac{\zeta \xi}{(\zeta + \xi)} \frac{\xi \eta}{(1 - \zeta)} \mathcal{G}_E \bar{\mathcal{G}}_E \sum_{\lambda_1} \sum_{l,r} (\delta^{lr} - 2i\epsilon^{lr} \lambda_1) (\delta^{lr} + 2i\epsilon^{lr} \lambda_1)^* = 0 \quad (3.43)$$

$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.44)$$

At large N_c limit, we get

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) &= Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) \\ S_{q\bar{q}g}(x, y, z) &= S(x, z) S(z, y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) &= S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) \end{aligned}$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - S(x, z) S(z, y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.45)$$

At large N_c limit, we get

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) &= Q(x', x', \bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) \\ S_{q\bar{q}g}(x', y, x') &= S(x', x') S(x', y) = S(x', y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) &= S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) \end{aligned}$$

$$\boxed{\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) - S(x', y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.46)$$

At large N_c limit, we get

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') &= Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) \\ S_{q\bar{q}g}(x, y, z) &= S(x, z) S(z, y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') &= S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) \end{aligned}$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) - S(x, z) S(z, y) - S(\bar{y}', \bar{x}) + 1}$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.47)$$

At large N_c limit, we get

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') &= Q(x', x', \bar{y}', \bar{x}) Q(x', y, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(x', y) \\ S_{q\bar{q}g}(x', y, x') &= S(x', x') S(x', y) = S(x', y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') &= S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) \end{aligned}$$

$$\boxed{\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(x', y) - S(x', y) - S(\bar{y}', \bar{x}) + 1}$$

A Corrections made to this version

For every item: [equation number(s) and page(s) in this PDF], then the line number(s) in the supplied `Trijet_massive.tex`. In the text, the changes are in red. Items 1–13 change results; items 14–30 are typos, notation or wording.

Errors that change results

1. [Eq. (3.1), p. 8] [Eq. (3.18), p. 11] [Eq. (3.39), p. 13] (L384, L493, L606): the factor Q^2 in the three cross-section prefactors was removed. With massive radial functions Q is already inside $\mathcal{G}, \mathcal{A}, \mathcal{G}_E$ (for $m_f = 0$, $\mathcal{G}^{(11)} \bar{\mathcal{G}}^{(11)} = \frac{1}{Y^2 \bar{Y}^2} \frac{Q^2 K_1(Q\mathcal{D}) K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}}$), so keeping Q^2 in front counts it twice.

2. [Eq. (3.1), p. 8; Eq. (3.18), p. 11; Eq. (3.39), p. 13] (same lines): $(\sum_f e_f^2)$ moved inside the coordinate integrals $(\sum_f e_f^2 [\dots])$, because every radial function depends on m_f and therefore on the flavour.
3. [Eq. (2.14), p. 4; Eq. (3.12), p. 9] [Eq. (2.37), p. 7; Eq. (3.29), p. 12] (L177, L428, L325, L539): $\mathcal{G}^{(00)}$, $\tilde{\mathcal{G}}^{(00)}$ had a spurious factor $\sqrt{A_x/\Delta_x}$. The general formula gives $(A_x/\Delta_x)^{(\alpha+\beta)/2} = 1$ for $\alpha = \beta = 0$, so $\mathcal{G}^{(00)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} K_0(\sqrt{A_x\Delta_x})$. Dimension check: $\mathcal{G}^{(00)}$ is dimensionless, while $\sqrt{A_x/\Delta_x}$ has dimension mass^2 .
4. [p. 3; p. 9] [p. 7; p. 11] (L163, L415, L311, L525): the prefactor of the general formula is a^β (resp. $\tilde{a}^\beta$), with $a = \frac{\xi\eta}{\zeta(1-\zeta)^2}$, not α^β (α is the helicity-path label). The argument of $\tilde{\mathcal{G}}$ is also $(\tilde{R}, X)$, not (R, X) .
5. [p. 5] (L214): the $\tilde{p}$ transform needs $e^{-i\tilde{p}\cdot Y}$ (the phase of L136) to give $-iY^n$, and the result is $-\frac{iY^n}{Y} \omega_f K_1(\omega_f Y)$, not $-\frac{iY^n}{Y^2} \omega_f K_1(\omega_f Y)$. The latter has the wrong dimension and does not reduce to $-iY^n/Y^2$ for $m_f \rightarrow 0$.
6. [Eq. (2.20), p. 5] (L217): $e^{+i\tilde{p}\cdot Y} \rightarrow e^{-i\tilde{p}\cdot Y}$, as in L136 and L156.
7. [Eq. (3.1), p. 8] [Eq. (3.1), p. 8] (L384): the Wilson-line arguments were wrong: $\mathcal{W}(x, y, y', \bar{x}, \bar{y}, \bar{z}) \rightarrow \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$ and $\mathcal{W}(x, y, y', \bar{x}, \bar{y}, \bar{y}') \rightarrow \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$. These are the structures actually worked out later (L459–L485).
8. [Eq. (3.3), p. 9] (L393): $F^{(02)} \rightarrow F^{(01)}$. There is no (02) path; this term is the square of the photon-flip path.
9. [Eq. (3.30), p. 12] (L543): the after-shock-wave vertex function of the *quark* channel was written with the antiquark structure $(\varphi(\zeta)\tau(\xi, \eta), R, Y, \omega_f)$. It must be $\tilde{\Psi}[\tilde{\mathcal{A}}]$, i.e. the structure of $\tilde{\Psi}[\tilde{\mathcal{G}}]$ with $\tilde{\mathcal{G}} \rightarrow \tilde{\mathcal{A}}$.
10. [Eq. (3.41), p. 13] (L615, interference, Regular $\times$ Regular, $\tilde{R} \cdot \bar{R}$ term): $\xi^2 \tilde{\omega}_f \omega_f \tilde{F}^{(01)} \bar{F}^{(10)} \rightarrow \xi^2 \tilde{\omega}_f \omega_f \tilde{F}^{(10)} \bar{F}^{(10)}$. Each gluon-flip vertex brings one $\xi\omega$; as printed the term had dimension mass^3 instead of mass^2 , and (01) does not go with the vector $\tilde{R}$.
11. [Eq. (1.2), p. 1] (L69): the antiquark term of the photon vertex lost its σ^λ ; it is $-\sigma^\lambda \frac{\sigma \cdot p + m_f \sigma^3}{p^+}$, as in L74.
12. [p. 2] (L101): middle expression $(\tilde{k}^2 + \omega_f^2) \rightarrow (\tilde{k}^2 + \epsilon_f^2)$.
13. [Eq. (2.24), p. 5] [Eq. (2.24), p. 5] (L248, momentum-space instantaneous term): the overall sign is $+$, not $-$, and the factor $\frac{1}{(2\pi)^6}$ and the colour matrix $t_{\alpha\beta}^a$ were missing. Only then does its Fourier transform give the coordinate-space expression of L256, whose $+$ sign is the correct one. The Fock state is written in the order $|q\bar{q}g\rangle$ as in the regular term.

Typos, notation and wording

14. [p. 4; p. 10] (L189, L444): $\epsilon_f = \tilde{Q}^2 + m_f^2 \rightarrow \epsilon_f^2 = \tilde{Q}^2 + m_f^2$.
15. [p. 1] (L63): $E_q(k) = \frac{k^2 + m^2}{2k^+} \rightarrow \frac{k^2 + m_f^2}{2k^+}$.
16. [p. 1] (L77): $\delta^{\lambda_1}, \delta^{\lambda_2} \rightarrow \delta^{\lambda,1}, \delta^{\lambda,2}$. These are the Cartesian components 1, 2 of the polarisation index λ , not the helicities.
17. [p. 2] (L86): the reduced gluon vertex was added. The text said that the p terms cancel but did not state the result, which defines the vertex of L98.
18. [p. 2; p. 4; p. 10] (L96, L188, L443): $\delta^{il} \rightarrow \delta^{i1}$ (letter $l \rightarrow$ number 1).
19. [Eq. (1.6), p. 2] (L103): missing “=”.
20. [Eq. (2.19), p. 4; Eq. (2.39), p. 7] (L206, L337): energy denominator $-E_q(k'') + E_q(p'') \rightarrow -E_q(k'') - E_q(p'')$.
21. [Eq. (2.21), p. 5] [Eq. (2.40), p. 8] (L228, L341): $\Psi_{\lambda_1\lambda}^{im}(\mathcal{A}) \rightarrow \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{A}]; [\tilde{\mathcal{A}}] \rightarrow [\tilde{\mathcal{A}}]$.
22. [p. 5] (L234–L235): “ $(b) = (a)|_{y,z \rightarrow y'}$ ” holds literally only for $m_f = 0$. The text now says what replaces it with masses, and that the $Y \rightarrow 0$ limits still agree.
23. [p. 6] (L258–L259): the sentence referred to the factor $\frac{Y^2}{R \cdot Y}$ of the massless effective vertex, which no longer appears; it was rephrased.
24. [p. 6] (L287): “=” $\rightarrow$ “ $\rightarrow$ ” in the rule for φ .
25. [p. 6] (L290): “ $D \rightarrow \tilde{Q}\tilde{R}$ ” is the massless after-shock-wave rule and uses the wrong photon scale for the quark channel. It was replaced by $\mathcal{D} \rightarrow \sqrt{\eta(1-\eta)} \tilde{R}$ (massless limit only; with masses, use $\tilde{\mathcal{A}}$).
26. [Eq. (3.5), p. 9] (L401): label “reg. $\times$ inst.” $\rightarrow$ “inst. $\times$ inst.”.
27. [Eq. (3.3), p. 9] [Eq. (3.20), p. 11] (L393, L502): $\bar{F}^{11} \rightarrow \bar{F}^{(11)}, \bar{F}^{01} \rightarrow \bar{F}^{(01)}$.
28. [Eq. (3.6), p. 9] [Eq. (3.7), p. 9] [Eq. (3.23), p. 11] [Eq. (3.24), p. 11] [Eq. (3.40), p. 13] (L406, L409, L515, L518, L610): $\mathcal{M}[\tilde{\mathcal{G}}] \equiv \mathcal{M}_1$ and $\mathcal{M}[\tilde{\mathcal{A}}] \equiv \mathcal{M}_2$ (and the same with tildes) are now stated, since $\mathcal{H}$ is defined through $\mathcal{M}[F]$. In $\mathcal{H}^x$: $\bar{\mathcal{M}}[\tilde{F}] \rightarrow \bar{\mathcal{M}}[\tilde{F}]$, because the bar is reserved for the conjugate-amplitude coordinates.
29. [p. 3] [p. 3] [p. 3; p. 7; p. 10; p. 12; p. 12] [p. 5] [p. 5]: spelling (“andf”, “is is”, “Explicitly”, “some normalisation”, “Instantaneous”).

Checked and correct as written: the final forms of the energy denominators, E_f^2 , the vertices φ, τ, μ , the momentum-space regular wave function, the Schwinger representation, $A_x, \Delta_x, \mathcal{G}^{(11)}, \mathcal{G}^{(10)}, \mathcal{G}^{(01)}$ and their tilde versions, $\mathcal{A}^{(\alpha\beta)}, \tilde{\mathcal{A}}^{(\alpha\beta)}, \mathcal{G}_E$, all coordinate-space wave functions except item 9, the effective vertices, all helicity sums except items 8 and 10, the overall signs ($-\zeta, +\eta$, interference $-\eta\zeta$), and all large- N_c Wilson-line structures.